

EE574 – Term Project

Due: Final Exam Date

1. Project Objective

The objective of this project is to design and implement a realistic state estimator for Energy Management System (EMS) using Python. Students will develop an AC Weighted Least Squares (WLS) state estimator using both conventional SCADA measurements and PMU phasor measurements. The project also includes observability analysis, and bad data detection/identification.

2. Project Scenario

A control center receives SCADA-type measurements (P/Q injections, P/Q flows, voltage magnitudes) every 5 seconds and PMU-type phasor measurements (voltage magnitude and angle) every 1 second at selected buses. Measurements may contain Gaussian noise, gross errors (outliers), missing PMU data, or insufficient coverage leading to observability issues. Your task is to build a state estimation pipeline capable of handling these realistic conditions.

3. Minimum Technical Requirements

- Implementation of AC-WLS State Estimation (iterative Newton-based approach).
- Integration of PMU measurements into the estimation model.
- Observability analysis (numerical method).
- Bad data detection and identification using normalized residual test.
- Comparison of at least three scenarios.

4. Deliverables

- Fully functional Python code (organized project structure).
- Single-command execution (e.g., `run_pipeline.py` or Jupyter notebook).
- Clear README explaining how to run the project.
- One-page scientific poster.
- Short demonstration.

5. Academic Integrity

External references and resources as well as use of AI must be clearly cited in the README file. Direct use of black-box state estimation solvers is not permitted.